Exactness-preserving discrete Maxwell operators on periodic polyhedral complexes

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Abstract

Problem. The standard Bloch-periodic extension of discrete exterior calculus (DEC) on unstructured polyhedral meshes breaks the discrete de Rham exact sequence: $d_1(\mathbf{k})\, d_0(\mathbf{k}) \neq 0$ for generic wave vector $\mathbf{k}$. This failure produces spectral pollution in the curl–curl eigenvalue problem, compromises the discrete Hodge splitting, and prevents mesh convergence of the acoustic dispersion relation. *Construction.* We build an exactness-preserving $d_1(\mathbf{k})$ via a face-boundary recurrence that derives the Bloch phases from those already present in the gradient $d_0(\mathbf{k})$. No per-edge phase assignment independent of $d_0(\mathbf{k})$ can achieve exactness on a generic unstructured mesh. The construction is unique up to a $U(1)$ phase per face, and the curl–curl operator $K(\mathbf{k}) = d_1(\mathbf{k})^\dagger \star_2 d_1(\mathbf{k})$ is canonical. *Results.* Verification on five periodic polyhedral complexes—including Voronoi foams and random meshes—confirms that exactness eliminates the pollution, restores the Hodge decomposition, and yields numerically second-order convergent ($O(h^2)$) acoustic dispersion with clean band structure across the full Brillouin zone.

Keywords: discrete exterior calculus, Bloch boundary conditions, Maxwell eigenvalues, polyhedral meshes, de Rham exactness, spectral pollution

1. Introduction

The discrete exterior calculus (DEC) provides a coordinate-free discretization of differential forms on cell complexes [14, 9]. For Maxwell's equations on periodic structures, DEC represents the electric field as a 1-cochain (edge values) and the magnetic flux as a 2-cochain (face values), with discrete gradient d_0 and curl d_1 acting between cochain spaces. The curl–curl eigenvalue

problem

$$K\,\mathbf{a} = \omega^2 M\,\mathbf{a}, \qquad K = d_1^\dagger \star_2 d_1, \qquad M = \star_1 \tag{1}$$

yields electromagnetic eigenfrequencies when the discrete de Rham sequence

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2 \tag{2}$$

is exact, i.e., $d_1\, d_0 = 0$. Boffi [3] established that exactness of the discrete complex is necessary and sufficient for spurious-free Maxwell eigenvalue computations: without it, the gauge kernel is incomplete and spectral pollution is unavoidable.

For periodic structures, Bloch boundary conditions twist the operators by a wave vector $\mathbf{k} \in \mathbb{R}^3$. On structured grids (Yee lattice), this is straightforward: the uniform topology ensures $d_1(\mathbf{k})\, d_0(\mathbf{k}) = 0$ automatically. However, on unstructured polyhedral meshes—as arise from Voronoi tessellations of periodic point sets—the standard construction of applying independent Bloch phases to each entry of d_1 breaks exactness at $\mathbf{k} \neq 0$. The core difficulty is not the twisting itself—twisted cochain complexes and flat local systems are classical [5]—but the incompatibility between independent edge-based phase lifting and the face-wise exactness constraints on non-tensorial meshes. On structured grids, the tensor-product topology ensures pairwise cancellation at every face. On unstructured meshes, the face boundaries mix edges with heterogeneous lattice shifts, and no per-edge phase assignment independent of $d_0(\mathbf{k})$ can restore exactness (Corollary 1).

The preservation of discrete exactness for Bloch-periodic DEC operators on general polyhedral meshes does not appear to have been treated explicitly in the literature. Existing DEC-based Maxwell computations use either structured grids where the issue does not arise [17], or non-periodic domains where Bloch conditions are unnecessary [7]. The discrete de Rham (DDR) method [10] achieves exactness on polyhedral meshes but has not been formulated with Bloch periodicity. The mimetic finite difference method of Jin, Xia, and Xu [15] treats Bloch boundary conditions on structured meshes but does not consider the unstructured case. Band structure computations on foam-like structures [16] use plane-wave expansion, not DEC.

Contribution. We present:

1. An explicit construction of an exactness-preserving $d_1(\mathbf{k})$ via a recurrence that derives the curl phases from those already present in $d_0(\mathbf{k})$, exploiting the flat holonomy of the Bloch connection on each face (§4).

2. A proof that $d_1(\mathbf{k})$ is unique up to a $U(1)$ phase per face, and that the curl–curl operator $K(\mathbf{k})$ is canonical (§4).
3. A systematic demonstration that the standard Bloch extension produces spectral pollution that enters the physical spectrum, destroys the Hodge splitting, and does not converge under mesh refinement (§5).
4. Numerical verification on five qualitatively different periodic polyhedral complexes, including random Voronoi meshes with no symmetry (§5–6).

Items (1)–(2) are mathematically new: the exactness failure of the standard Bloch extension on unstructured meshes has not, to our knowledge, been previously identified, and the face-boundary recurrence that corrects it is a new construction. Items (3)–(4) provide the first numerical evidence that this structural error produces spectral pollution, broken Hodge splitting, and convergence failure on realistic foam geometries.

2. Periodic complexes and DEC operators

2.1. Cell complex on the torus

Let $\Lambda = L_x\mathbb{Z} \times L_y\mathbb{Z} \times L_z\mathbb{Z}$ be a lattice in $\mathbb{R}^3$ with fundamental domain $\Omega = [0, L_x) \times [0, L_y) \times [0, L_z)$. A periodic cell complex (V, E, F, C) on the flat torus $T^3 = \mathbb{R}^3/\Lambda$ consists of vertices V, oriented edges E, oriented faces F, and 3-cells C.

Edges connecting vertices across the periodic boundary carry a lattice shift vector $\mathbf{n}_e \in \mathbb{Z}^3$:

$$\mathbf{n}_e = \mathrm{round}\big((\mathbf{x}_j - \mathbf{x}_i)/\mathbf{L}\big) \tag{3}$$

where division is componentwise. Interior edges have $\mathbf{n}_e = \mathbf{0}$.

2.2. Incidence matrices

The topological gradient $d_0 \in \mathbb{R}^{|E|\times|V|}$ and curl $d_1 \in \mathbb{R}^{|F|\times|E|}$ encode the boundary relations:

$$(d_0)_{ev} = \begin{cases} +1 & \text{if } v = \mathrm{head}(e), \\ -1 & \text{if } v = \mathrm{tail}(e), \\ 0 & \text{otherwise}, \end{cases} \tag{4}$$

$$(d_1)_{fe} = \begin{cases} +1 & \text{if } e \in \partial f, \text{ same orientation}, \\ -1 & \text{if } e \in \partial f, \text{ opposite orientation}, \\ 0 & \text{if } e \notin \partial f. \end{cases} \tag{5}$$

These satisfy $d_1 d_0 = 0$ (every vertex appears in the boundary of a face an even number of times with canceling signs).

2.3. Hodge stars

The diagonal Hodge star operators $\star_1 \in \mathbb{R}^{|E|\times|E|}$ and $\star_2 \in \mathbb{R}^{|F|\times|F|}$ encode metric information from the dual complex:

$$(\star_1)_{ee} = \frac{|\sigma_e^*|}{|\sigma_e|}, \qquad (\star_2)_{ff} = \frac{|\sigma_f^*|}{|\sigma_f|} \tag{6}$$

where σ_e, σ_f are primal edge length and face area, and σ_e^*, σ_f^* are their dual counterparts. For the SC cubic benchmark with lattice constant a, all stars are uniform: $\star_1 = a\,I$, $\star_2 = a\,I$.

2.4. Test structures

We verify the construction on five periodic polyhedral complexes spanning three qualitatively different categories.

Structure	Type	$\lvert V\rvert$	$\lvert E\rvert$	$\lvert F\rvert$	Plateau	O_h
SC cubic (N=3)	regular lattice	27	81	81	no	yes
C15 (Laves)	Voronoi foam	136	272	160	yes	yes
Kelvin (BCC)	Voronoi foam	96	192	112	yes	yes
Weaire–Phelan	Voronoi foam	46	92	54	yes	yes
Random Voronoi	random	206–354	412–708	236–404	yes	no

Table 1: Test structures. Plateau foams have 4 edges per vertex and 3 faces per edge. The random Voronoi complexes are constructed from 50 uniformly distributed points; the range reflects 10 different seeds.

3. The exactness problem

3.1. Bloch-twisted gradient

For wave vector $\mathbf{k} \in \mathbb{R}^3$, the Bloch-twisted gradient is (where $\mathbf{n}_e\mathbf{L}$ denotes the componentwise product $(n_{e,1}L_x,\, n_{e,2}L_y,\, n_{e,3}L_z)$):

$$d_0(\mathbf{k})[e,v] = \begin{cases} +e^{i\mathbf{k}\cdot\mathbf{n}_e\mathbf{L}} & \text{if } v = \text{head}(e), \\ -1 & \text{if } v = \text{tail}(e), \\ \ \ 0 & \text{otherwise.} \end{cases} \tag{7}$$

This is standard and well-defined: each edge carries a unique lattice shift.

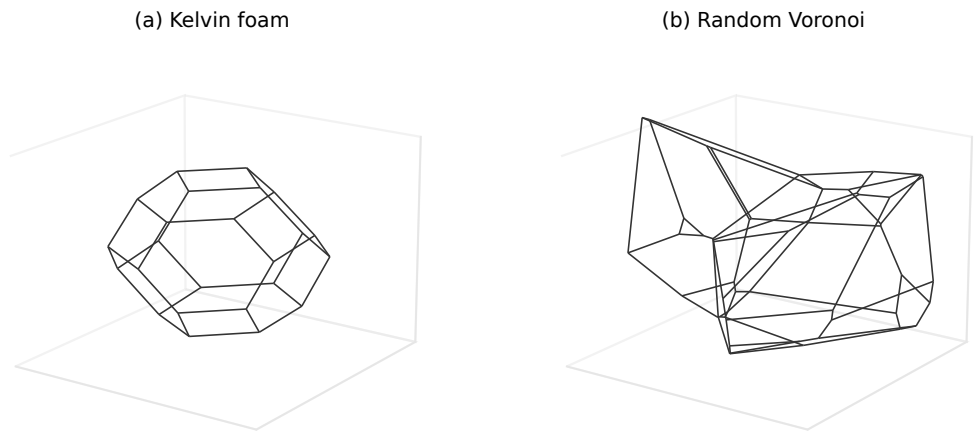

Figure 1: Wireframe view of two test structures (central region of the periodic box). (a) Kelvin foam: regular BCC Voronoi tessellation producing truncated octahedra with hexagonal and square faces. (b) Random Voronoi: irregular polyhedral cells with no lattice symmetry. The full periodic complexes used in the computations contain $|V| = 96$–354 vertices (Table 1).

3.2. Standard Bloch curl

The standard extension applies independent Bloch phases per face:

$$d_1^{\mathrm{std}}(\mathbf{k})[f, e] = d_1[f, e] \cdot e^{i\mathbf{k}\cdot\mathbf{n}_e\mathbf{L}}. \tag{8}$$

On tensor-product (structured) grids, this preserves exactness: each face is a quadrilateral whose opposite edges carry identical lattice shifts, so the Bloch phases cancel pairwise at every vertex. On unstructured meshes, faces generically have boundary edges with heterogeneous shifts, and exactness fails.

3.3. Diagnosis

The product $d_1(\mathbf{k})\, d_0(\mathbf{k})$ at a face f and vertex v yields

$$(d_1 d_0)[f, v] = \sum_{\substack{e \in \partial f \\ v \in e}} d_1(\mathbf{k})[f, e] \cdot d_0(\mathbf{k})[e, v]. \tag{9}$$

At $\mathbf{k} = \mathbf{0}$, cancellation is guaranteed by the topological constraint $d_1 d_0 = 0$. At $\mathbf{k} \neq \mathbf{0}$, the phases on different boundary edges prevent cancellation unless they are mutually consistent.

Proposition 1 (Standard construction breaks exactness)**.** *Let f be a face with at least two boundary edges e_a, e_b sharing a vertex and having distinct lattice shifts $\mathbf{n}_a \neq \mathbf{n}_b$. Then $d_1^{\mathrm{std}}(\mathbf{k})\, d_0(\mathbf{k}) \neq 0$ for almost all $\mathbf{k} \in \mathbb{R}^3$.*

Proof. At the shared vertex v, the two contributions carry phases $e^{i\mathbf{k}\cdot\mathbf{n}_a\mathbf{L}}$ and $e^{i\mathbf{k}\cdot\mathbf{n}_b\mathbf{L}}$. Cancellation requires $e^{i\mathbf{k}\cdot(\mathbf{n}_a-\mathbf{n}_b)\mathbf{L}} = 1$, which defines a hyperplane in $\mathbf{k}$-space. This holds at $\mathbf{k} = \mathbf{0}$ but fails generically when $\mathbf{n}_a \neq \mathbf{n}_b$. □

More generally, the failure is not specific to the standard phase choice:

Corollary 1 (No per-edge phase assignment suffices)**.** *Let $\varphi_e(\mathbf{k}) \in \mathbb{C}^*$ be any phase assigned to edge e, depending only on $\mathbf{k}$ and the lattice shift $\mathbf{n}_e$ (not on $d_0(\mathbf{k})$). Define $\tilde{d}_1(\mathbf{k})[f, e] = d_1[f, e] \cdot \varphi_e(\mathbf{k})$. Then $\tilde{d}_1(\mathbf{k})\, d_0(\mathbf{k}) \neq 0$ for generic $\mathbf{k}$ on any complex satisfying the hypotheses of Proposition 1.*

Proof. At vertex v_i shared by consecutive boundary edges e_{i-1}, e_i of face f, exactness requires

$$\varphi_i/\varphi_{i-1} = -d_0(\mathbf{k})[e_{i-1}, v_i] \,/\, d_0(\mathbf{k})[e_i, v_i]. \tag{10}$$

By hypothesis φ_e depends only on $\mathbf{n}_e$. Consider an edge e that appears in two faces f_1 and f_2 with different predecessors e' (in f_1) and e'' (in f_2), sharing vertices v' and v'' respectively. The constraint (10) requires

$$\varphi(\mathbf{n}_e)/\varphi(\mathbf{n}_{e'}) = R_1(\mathbf{k}), \qquad \varphi(\mathbf{n}_e)/\varphi(\mathbf{n}_{e''}) = R_2(\mathbf{k}),$$

where R_1, R_2 are ratios of entries of $d_0(\mathbf{k})$ evaluated at different vertices with different lattice shifts. On an unstructured mesh satisfying the hypotheses of Proposition 1, there exist such pairs with $R_1(\mathbf{k}) \cdot \varphi(\mathbf{n}_{e'}) \neq R_2(\mathbf{k}) \cdot \varphi(\mathbf{n}_{e''})$ for generic $\mathbf{k}$, since R_1/R_2 involves exponentials $e^{i\mathbf{k}\cdot(\mathbf{n}_{e'}-\mathbf{n}_{e''})\mathbf{L}}$ that are non-constant when $\mathbf{n}_{e'} \neq \mathbf{n}_{e''}$. □

The hypothesis—an edge shared by two faces with predecessor edges carrying distinct lattice shifts—holds on every non-tensorial polyhedral complex with boundary-crossing faces, including all Voronoi foams and random meshes tested here.

3.4. Quantification

The exactness violation $\|d_1^{\mathrm{std}}(\mathbf{k})\, d_0(\mathbf{k})\|$ is not a small perturbation. On the Kelvin foam at 10% of the Brillouin zone:

$$\|d_1^{\mathrm{std}}\, d_0\| = 7.5, \qquad \|d_1^{\mathrm{exact}}\, d_0\| = 8 \times 10^{-16}. \tag{11}$$

The violation grows with $|\mathbf{k}|$ and is present on all tested unstructured meshes (Table 5).

4. Exactness-preserving construction

4.1. Flat holonomy

Lemma 1 (Flat holonomy)**.** *Let f be a face with boundary vertices $v_0, v_1, \ldots, v_{n-1}$ and boundary edges $e_0, \ldots, e_{n-1}$ where e_i connects v_i to $v_{(i+1) \bmod n}$. Define the holonomy*

$$H_f = \prod_{i=0}^{n-1} \frac{-d_0(\mathbf{k})[e_i, v_{i+1}]}{d_0(\mathbf{k})[e_i, v_i]}. \tag{12}$$

Then $H_f = 1$ for all $\mathbf{k}$.

Proof. Each factor equals $e^{i\mathbf{k}\cdot\mathbf{n}_i\mathbf{L}}$. Therefore $H_f = \exp(i\mathbf{k} \cdot (\sum_i \mathbf{n}_i)\mathbf{L})$. Each face is a 2-cell in the periodic CW-decomposition of T^3 induced by a tessellation of $\mathbb{R}^3$; it lifts to a contractible polygon in $\mathbb{R}^3$, so its boundary carries zero net lattice translation: $\sum_i \mathbf{n}_i = \mathbf{0}$, and $H_f = 1$. □

4.2. Recurrence construction

Theorem 1 (Exactness-preserving d_1)**.** *There exists a linear operator $d_1(\mathbf{k})\colon C^1 \to C^2$ satisfying $d_1(\mathbf{k})\, d_0(\mathbf{k}) = 0$ for all $\mathbf{k}$. The construction is explicit: for each face f with ordered boundary edges $e_0, \ldots, e_{n-1}$ and incidence signs $\sigma_i = \pm 1$:*

1. *Set $\varphi_0 = 1$.*
2. *For $i = 1, \ldots, n-1$:*

$$\varphi_i = -\sigma_{i-1}\, \varphi_{i-1}\, \frac{d_0(\mathbf{k})[e_{i-1}, v_i]}{\sigma_i \cdot d_0(\mathbf{k})[e_i, v_i]}. \tag{13}$$

3. *Set $d_1(\mathbf{k})[f, e_i] = \sigma_i\, \varphi_i$.*

Proof. The condition $d_1(\mathbf{k})\, d_0(\mathbf{k}) = 0$ at face f decomposes into n vertex equations. At vertex v_i for $i = 1, \ldots, n-1$:

$$d_1[f, e_{i-1}] \cdot d_0(\mathbf{k})[e_{i-1}, v_i] + d_1[f, e_i] \cdot d_0(\mathbf{k})[e_i, v_i] = 0. \tag{14}$$

Substituting $d_1[f, e_i] = \sigma_i \varphi_i$ yields the recurrence (13). The remaining equation at v_0 holds if and only if $H_f = 1$, which is guaranteed by Lemma 1. □

The construction requires one pass over the boundary edges of each face. Its computational cost is $O(\sum_f n_f)$, where n_f is the number of edges of face f—the same as assembling the standard d_1. The only additional input is $d_0(\mathbf{k})$, which is already available.

4.3. Uniqueness

Corollary 2 (Uniqueness)**.** *Any local operator $\tilde{d}_1(\mathbf{k})$ satisfying (i) $\tilde{d}_1\, d_0 = 0$ and (ii) $\tilde{d}_1[f, e] = 0$ whenever $e \notin \partial f$ differs from $d_1(\mathbf{k})$ by a scalar $\lambda_f \in \mathbb{C}^*$ per face: $\tilde{d}_1[f, :] = \lambda_f\, d_1[f, :]$.*

Proof. The exactness condition at face f is a linear recurrence of order 1 with $n - 1$ equations in n unknowns. Once $\tilde{d}_1[f, e_0]$ is chosen (the seed), all other entries are determined by (13). Therefore the solution space is one-dimensional per face. □

Remark 1. *Since each recurrence step is a ratio of entries of $d_0(\mathbf{k})$—which have unit modulus—all phases satisfy $|\varphi_i| = 1$. The entries of $d_1(\mathbf{k})$ are therefore unimodular: $|d_1(\mathbf{k})[f, e]| = 1$ for all $e \in \partial f$.*

4.4. Canonical curl–curl operator

Proposition 2 (Canonical K)**.** *Let $d_1(\mathbf{k})$ and $\tilde{d}_1(\mathbf{k})$ be two operators constructed by Theorem 1, differing in starting vertex, cyclic orientation, or initial phase seed. Then*

$$\tilde{d}_1(\mathbf{k})^\dagger \star_2 \tilde{d}_1(\mathbf{k}) = d_1(\mathbf{k})^\dagger \star_2 d_1(\mathbf{k}). \tag{15}$$

Proof. By Corollary 2, any two constructions differ by $\tilde{d}_1[f, :] = \lambda_f\, d_1[f, :]$ with $|\lambda_f| = 1$. Since $K = \sum_f \star_2[f, f] \cdot d_1[f, :]^\dagger d_1[f, :]$, each term acquires $|\lambda_f|^2 = 1$. □

Numerical confirmation. On all test structures, six permutation types (cyclic shifts, orientation reversal, random permutations): $\|\Delta K\|_F < 10^{-14}$, while $\|\Delta d_1\|_F \sim 10^1$.

Remark 2 (Relation to the standard construction)**.** *The exactness-preserving $d_1(\mathbf{k})$ is not a diagonal gauge transform of $d_1^{\mathrm{std}}(\mathbf{k})$. If $d_1(\mathbf{k}) = D\, d_1^{\mathrm{std}}(\mathbf{k})$ for some diagonal unitary D, then d_1^{std} would also satisfy $d_1^{\mathrm{std}}(\mathbf{k})\, d_0(\mathbf{k}) = 0$, contradicting Proposition 1. The two operators differ structurally, not merely by a choice of gauge.*

4.5. Gauge kernel

Proposition 3 (Kernel dimension)**.** *For generic $\mathbf{k}$ (i.e., $e^{ik_jL_j} \neq 1$ for $j = 1, 2, 3$),* $\dim\ker K(\mathbf{k}) = |V|$.

Proof. We proceed in three algebraic steps.

Step 1: Injectivity of $d_0(\mathbf{k})$. If $d_0(\mathbf{k})\mathbf{u} = \mathbf{0}$, then for every edge (i, j): $u_j \cdot e^{i\mathbf{k}\cdot\mathbf{n}_e\mathbf{L}} = u_i$. Iterating along a closed path around the j-th torus cycle requires $u_i \cdot e^{ik_jL_j} = u_i$. For generic $\mathbf{k}$ ($e^{ik_jL_j} \neq 1$), this forces $u_i = 0$. Hence $d_0(\mathbf{k})$ is injective and $\dim \operatorname{im} d_0(\mathbf{k}) = |V|$.

Step 2: Lower bound. Since $\star_2$ is positive definite, $\ker K = \ker d_1(\mathbf{k})$. Exactness gives $\operatorname{im} d_0(\mathbf{k}) \subseteq \ker d_1(\mathbf{k})$, so $\dim \ker K(\mathbf{k}) \geq |V|$.

Step 3: Upper bound. The periodic cell complex (V, E, F, C) is a CW-decomposition of T^3, and the Bloch twist equips the cellular cochain complex with coefficients in the rank-one local system $\mathcal{L}_\mathbf{k}$ determined by the representation $\pi_1(T^3) \cong \mathbb{Z}^3 \to U(1)$ with holonomy $e^{ik_jL_j}$ along the j-th cycle. By the standard identification of cellular and singular cohomology with local coefficients, this complex computes $H^p(T^3, \mathcal{L}_\mathbf{k})$. Since $T^3 = S^1 \times S^1 \times S^1$ and the cohomology of S^1 with coefficients in a nontrivial local system vanishes (see e.g. [5], Ch. I), the Künneth formula gives $H^p(T^3, \mathcal{L}_\mathbf{k}) = 0$ for all p when $e^{ik_jL_j} \neq 1$ for all j. In particular $H^1 = 0$, so $\ker d_1(\mathbf{k}) = \operatorname{im} d_0(\mathbf{k})$, giving $\dim \ker K(\mathbf{k}) = |V|$. □

Numerical confirmation. On all test structures (Table 1), for 20+ wave vectors spanning the Brillouin zone interior, $\operatorname{rank}(d_1(\mathbf{k})) = |E| - |V|$ to machine precision.

5. Spectral consequences

We now demonstrate the spectral impact of exactness failure. All results compare the standard construction (8) with the exactness-preserving construction of Theorem 1 on the same mesh and wave vector.

5.1. Spectral pollution

The exactness-preserving $d_1(\mathbf{k})$ produces a gauge kernel of dimension $|V|$, consistent with Proposition 3. The standard $d_1^{\mathrm{std}}(\mathbf{k})$ produces a smaller kernel: modes that should be gauge zeros are expelled into the physical spectrum as spurious eigenvalues.

The spurious eigenvalues are not isolated—they fall inside the physical band and are indistinguishable from genuine physical modes on the basis of eigenvalue alone. On the Kelvin foam, the 6 spurious eigenvalues (axis-aligned $\mathbf{k}$) have $\omega^2 \in [0.026, 0.146]$, fully overlapping with the physical spectrum.

Structure	[100] direction			generic direction		
	$n_{\text{zero}}^{\text{exact}}$	$n_{\text{zero}}^{\text{std}}$	n_{spur}	$n_{\text{zero}}^{\text{exact}}$	$n_{\text{zero}}^{\text{std}}$	n_{spur}
Kelvin ($\lvert V\rvert$=96)	96	90	6	96	82	14
C15 ($\lvert V\rvert$=136)	136	127	9	136	120	16
WP ($\lvert V\rvert$=46)	46	43	3	46	39	7
Random ($\lvert V\rvert\approx$340)	$\lvert V\rvert$	$\lvert V\rvert-15$	12–20	—	—	—

Table 2: Gauge kernel dimension at 10% of the Brillouin zone. The standard construction produces fewer gauge zeros; the missing modes appear as spurious eigenvalues inside the physical spectrum. The number of spurious modes depends on the **k**-direction: generic directions produce more spurious modes than axis-aligned directions. (Random Voronoi: generic direction not tested; the lack of lattice symmetry makes the [100] result already representative of a generic direction.)

The number of spurious modes is direction-dependent: axis-aligned **k**-vectors produce fewer spurious modes than generic directions (Table 2). This suggests that benchmarks restricted to high-symmetry directions may underestimate the extent of the pollution.

5.2. Mesh convergence

The acoustic wave speed $c^2 = \omega^2/|\mathbf{k}|^2$ extracted from the lowest physical eigenvalue should converge to a mesh-independent limit under mesh refinement. Table 3 shows the result for the Kelvin foam at 5% of the Brillouin zone along [100]. (The Kelvin foam admits a natural refinement family: N BCC unit cells per axis, giving $|V| = 96N^3/8$. The other test structures have fixed unit cells and do not admit comparable parametric refinement.) In both the exact and standard constructions, the "physical" eigenvalues are identified by the same criterion: all eigenvalues with $|\omega^2| < \epsilon_{\text{rel}} \cdot \max|\omega^2|$ (with $\epsilon_{\text{rel}} = 10^{-12}$) are classified as gauge zeros, and the remaining eigenvalues are sorted in ascending order. No pre-selection or filtering is applied.

The exactness-preserving construction yields c^2 that converges monotonically to the correct continuum value. The error $|1-c^2|$ decreases as $O(1/N^2)$ over the tested range (Figure 2a), consistent with second-order convergence typical of lowest-order structure-preserving discretizations on exact subcomplexes [1]. The standard construction oscillates with increasing amplitude ($|\Delta c^2_{\text{std}}|$ grows from 0.5 to 2.3 over the tested range) and does not exhibit convergence under mesh refinement. The exact construction also preserves

N	$\lvert V\rvert$	c^2_{exact}	c^2_{std}	$\lvert 1 - c^2_{\text{exact}}\rvert$	$\lvert \Delta c^2_{\text{std}}\rvert$
2	96	0.99968	1.549	3.2×10^{-4}	—
3	324	0.99986	2.476	1.4×10^{-4}	9.3×10^{-1}
4	768	0.99992	1.933	8.0×10^{-5}	5.4×10^{-1}
5	1500	0.99995	1.029	5.1×10^{-5}	9.0×10^{-1}
6	2592	0.99996	3.341	3.6×10^{-5}	2.3×10^{0}

Table 3: Acoustic wave speed under mesh refinement (Kelvin foam, 5% BZ along [100], mesh spacing $h = L/N$). The exactness-preserving construction converges monotonically to $c^2 = 1$ with observed second-order rate: the error ratios $|1 - c^2_N|/|1 - c^2_{N+1}|$ match $(N{+}1)^2/N^2$ to three significant figures. The standard construction oscillates ($1.55 \to 2.48 \to 1.93 \to 1.03 \to 3.34$) with non-monotonic $|\Delta c^2|$.

acoustic degeneracy (modes 0 and 1 identical to machine precision), while the standard construction breaks it at all refinement levels.

5.3. Hodge splitting

The discrete Hodge decomposition of 1-forms into gradient and curl subspaces requires $\mathrm{im}(d_0) \perp \ker(d_1^\dagger \star_2)$, which follows from $d_1\, d_0 = 0$. We test this by computing the full Hodge Laplacian $\Delta_1 = K_{\text{grad}} + K_{\text{curl}}$ and projecting each eigenvector onto $\mathrm{im}(d_0)$.

	Exact $d_1(\mathbf{k})$	Standard $d_1^{\text{std}}(\mathbf{k})$
Pure gradient modes	96	5–23
Pure curl modes	96	0–7
Mixed modes	0	162–187
Total	192	192

Table 4: Hodge splitting of the full Hodge Laplacian eigenvectors on the Kelvin foam ($|E|$=192) at 5–10% BZ. A mode is classified as "pure" if its projection onto $\mathrm{im}(d_0)$ is > 0.99 or < 0.01, and "mixed" otherwise. The exactness-preserving construction achieves perfect separation; the standard construction mixes 85–97% of modes.

With the exactness-preserving $d_1(\mathbf{k})$, every eigenvector of Δ_1 is either purely gradient ($\|P_{\text{grad}}\mathbf{u}\|/\|\mathbf{u}\| = 1.000$) or purely curl ($= 0.000$), with zero mixed modes out of 192. The standard construction produces 85–97% mixed modes—nearly every eigenvector is a gradient–curl mixture.

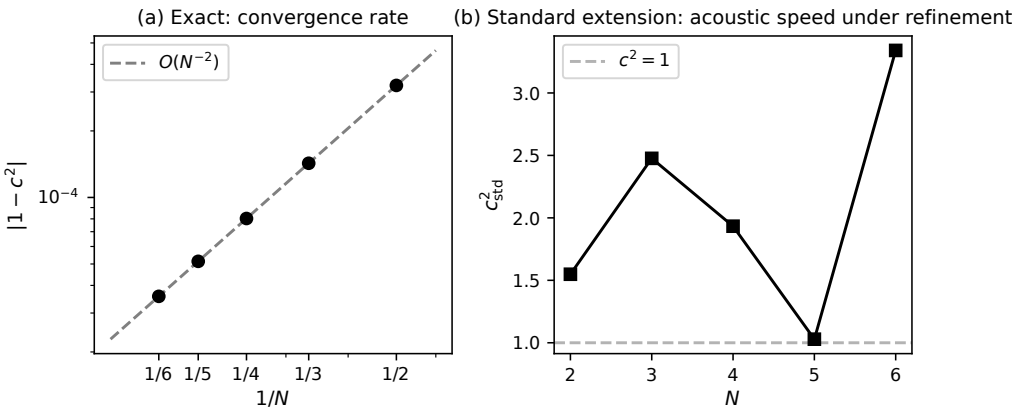

Figure 2: Mesh convergence of the acoustic wave speed (Kelvin foam, 5% BZ along [100]). (a) Exactness-preserving construction: the error $|1 - c^2|$ decreases as $O(1/N^2)$ on a log–log plot ($p = 2.00$, dashed line). (b) Standard construction: c^2_{std} oscillates with increasing amplitude under refinement and does not approach the continuum value $c^2 = 1$ (dashed line).

This explains the spectral pollution structurally: in the absence of exactness, gradient modes are no longer annihilated by the curl operator and acquire a curl component through the broken Hodge splitting, making them visible to the curl–curl eigenvalue problem as spurious modes.

5.4. Band structure

Figure 3 shows the full band structure along the standard cubic Brillouin zone path Γ–X–M–R–Γ for the Kelvin foam (N=2, $|V|$=96, $|E|$=192). The exactness-preserving construction (a) produces clean, well-separated bands with the expected twofold acoustic degeneracy at Γ (blue lines). The standard construction (b) produces dense spurious bands that collapse toward zero near every high-symmetry point, rendering the band structure severely contaminated. At Γ itself ($\mathbf{k} = \mathbf{0}$), both constructions coincide because $d_1^{\mathrm{std}}(\mathbf{0}) = d_1$; the pollution appears as soon as $\mathbf{k} \neq \mathbf{0}$. Panel (c) confirms that the failure is algebraic, not numerical: $\|d_1^{\mathrm{std}}(\mathbf{k})\, d_0(\mathbf{k})\| \sim 10^1$ everywhere along the path, while $\|d_1^{\mathrm{exact}}(\mathbf{k})\, d_0(\mathbf{k})\| \sim 10^{-16}$ (machine precision).

5.5. Universality

The phenomena described above are universal across all tested structures: Three additional observations:

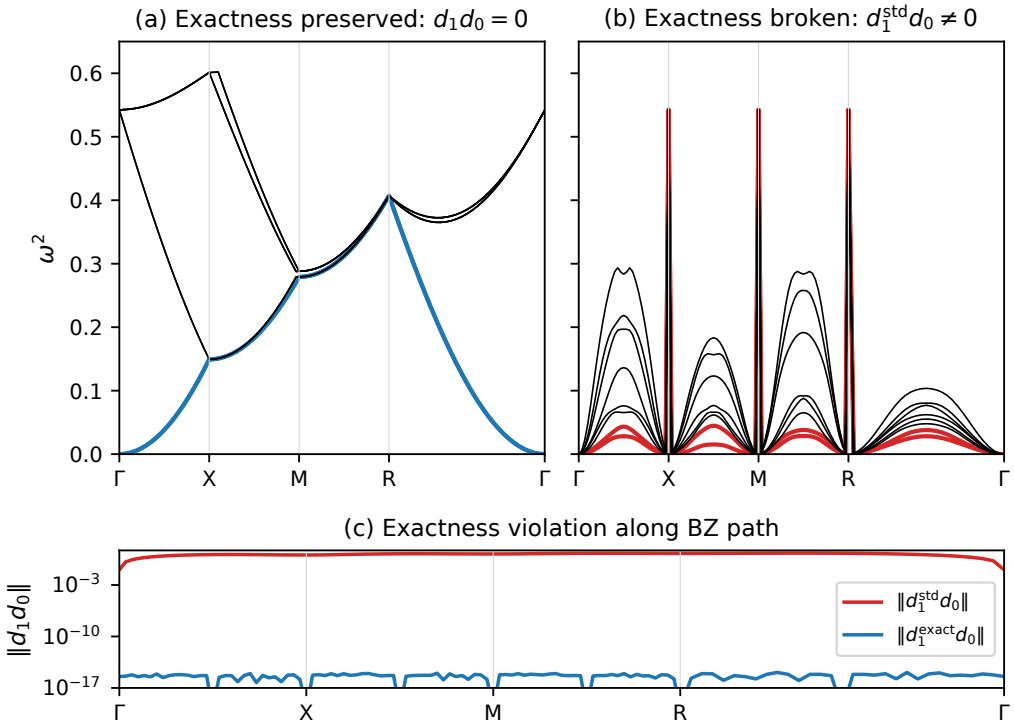

Figure 3: Band structure along Γ–X–M–R–Γ for the Kelvin foam (N=2, $|V|$=96, $|E|$=192, 121 **k**-points). (a) Exactness-preserving $d_1(\mathbf{k})$: clean bands with twofold acoustic degeneracy (blue). (b) Standard $d_1^{\mathrm{std}}(\mathbf{k})$: spurious bands (red, black) collapse toward zero; band structure is severely contaminated. (c) Exactness violation $\|d_1\, d_0\|$ along the path: the standard construction has $O(10^1)$ violation (red), while the exact construction maintains $\|d_1\, d_0\| < 10^{-15}$ (blue) at every **k**-point.

- *Extensive pollution.* The number of spurious modes grows with system size: $n_{\mathrm{spur}} = 6$ at N=2 ($|V|$=96), 17 at N=3 ($|V|$=324) for the Kelvin foam along [100]. The pollution is not a fixed topological defect but scales with the mesh.

- *Specific to cochain level* 1→2. The scalar Laplacian $K_0 = d_0^\dagger \star_1 d_0$ (acting on 0-forms) has a clean spectrum with no spurious modes. The gradient $d_0(\mathbf{k})$ is uniquely defined and involves no exactness ambiguity. The pollution arises specifically at the curl level, where $d_1(\mathbf{k})$ must satisfy $d_1 d_0 = 0$.

- *Onset at minimal supercell.* On the minimal SC cubic complex ($|V|$=1,

Structure	$\|d_1^{\text{std}} d_0\|$	n_{spur}	c^2_{exact}	c^2_{std}	Leakage
SC N=3	6.2	5	1.000	—	—
Kelvin N=2	7.5	6–14	0.999	1.48	0.88
C15 N=1	7.8	9–16	0.999	0.52	0.91
WP N=1	5.1	3–7	0.999	1.20	0.92
Random (10 seeds)	9–12	12–20	0.998	0.43–1.06	0.88–0.91

Table 5: Universality of exactness failure and its consequences at 10% BZ. The column "Leakage" gives the mean gradient projection $\|P_{\text{grad}}\mathbf{u}\|/\|\mathbf{u}\|$ of the spurious modes in the standard construction. All exactness-preserving results have $\|d_1 d_0\| < 10^{-15}$, $n_{\text{spur}} = 0$, and leakage = 0. (SC cubic: c^2_{std} and leakage omitted because the uniform Hodge stars on a regular lattice make $c^2_{\text{exact}} = 1$ by symmetry, not by convergence; the metric quantities are not directly comparable with those on the Voronoi foams.)

$|E|$=3, $|F|$=3), each face is a square with two pairs of opposite edges carrying equal lattice shifts—a tensor-product topology where pairwise cancellation is automatic. The standard phase assignment therefore preserves exactness, and $d_1^{\text{std}}(\mathbf{k})\, d_0(\mathbf{k}) = 0$. Pollution appears at the first non-trivial supercell (SC N=3: 5 spurious modes), where faces acquire boundary edges with heterogeneous shifts.

6. Robustness

We classify perturbations into five structurally independent categories and test each on the Kelvin foam (N=2). In addition to these targeted perturbations, the spectral consequences of §5 were verified under: random $\mathbf{k}$-directions (23 directions including high-symmetry and generic orientations), $\mathbf{k} \to \mathbf{0}$ asymptotics (10^{-4} to 20% BZ), Brillouin zone boundary behavior (up to 99.9% BZ), system size scaling ($N = 2, 3, 4$), 10 random Voronoi topologies, and the operator norm $\|K_{\text{std}} - K_{\text{exact}}\|_F$ as a function of $|\mathbf{k}|$. In all cases, the exactness-preserving construction produces the correct kernel dimension and convergent acoustic spectrum. The failure of the standard construction is structural: it persists under refinement, random topology, random $\mathbf{k}$-directions, metric perturbations, and $\mathbf{k} \to 0$ asymptotics, and is not an artifact of insufficient resolution, special symmetry, or particular mesh choice.

Table 6 summarizes the five targeted perturbation types. Exactness and kernel dimension are topological invariants; wave speed and degeneracy are

metric and symmetry properties.

Perturbation	Range	$d_1 d_0 = 0$	$\ker = \lvert V \rvert$	c^2, degeneracy
Geometric (vertex positions)	$\varepsilon = 0$–0.10	yes	yes	c^2 stable, split breaks
Threshold sensitivity	10^{-8}–10^{-14}	yes	yes	unchanged
BZ boundary ($\mathbf{k} = \mathbf{k}_{\mathrm{BZ}}$)	0.1–0.999 BZ	yes	yes	$c^2 \to 0$ (band edge)
Face ordering (6 permutations)	all	yes	yes	K identical
Gauge transform (random $\theta(v)$)	5 seeds	yes	yes	eigenvalues identical

Table 6: Robustness summary on the Kelvin foam. Exactness and kernel dimension are preserved under all perturbation types. At the exact BZ boundary ($\mathbf{k} = \mathbf{k}_{\mathrm{BZ}}$), $\mathrm{rank}(d_0)$ drops by 1 and the kernel gains two additional zeros, but exactness is maintained and no numerical instability occurs.

Property	Type	Robust under
$d_1(\mathbf{k})\, d_0(\mathbf{k}) = 0$	topological	all perturbations
$\dim \ker K = \lvert V \rvert$	topological	all perturbations
$\omega^2 = c^2 \lvert \mathbf{k} \rvert^2$	metric	geometry, symmetry
Polarization degeneracy	symmetry	topology, metric

Table 7: Classification of spectral properties.

7. Discussion

Relation to Boffi's exactness criterion. Boffi [3] showed that exactness of the discrete de Rham complex is necessary and sufficient for spurious-free Maxwell eigenvalue computations; earlier work by Boffi, Conforti, and Gastaldi [4] demonstrated this in the specific context of photonic crystal eigenproblems, and Dobson and Pasciak [11] analyzed related algorithms for electromagnetic Bloch modes. Our results provide a concrete demonstration: the standard Bloch extension violates exactness on unstructured meshes, and the resulting spectral pollution, broken Hodge splitting, and convergence failure are precisely the consequences predicted by Boffi's theory. The exactness-preserving construction restores all three.

Connection interpretation. The exactness violation $\|d_1^{\mathrm{std}}(\mathbf{k})\, d_0(\mathbf{k})\|$ can be interpreted as the curvature of a discrete connection. The standard Bloch extension assigns phases independently per edge, defining a connection with $O(1)$ curvature that grows with $|\mathbf{k}|$. The curvature manifests algebraically as the non-vanishing of $d_1^{\mathrm{std}}(\mathbf{k})\, d_0(\mathbf{k})$. The recurrence construction of Theorem 1 produces a flat connection (zero curvature) by deriving all phases from those in $d_0(\mathbf{k})$. This interpretation connects our construction to the framework of bundle-valued DEC [6].

Existing discretizations. The finite element exterior calculus (FEEC) [1, 2] and its smoothed projection framework [8] provide the theoretical foundation for structure-preserving Maxwell discretizations; Hiptmair [13] gives a comprehensive treatment of edge elements in this context. The discrete de Rham (DDR) method [10] achieves exactness on polyhedral meshes, and the topological framework of Gross and Kotiuga [12] treats electromagnetism on complexes. The mimetic finite difference method of [15] treats Bloch conditions on structured meshes. None of these, however, addresses the specific setting of our work: Bloch-periodic DEC on unstructured polyhedral meshes. FEEC does not treat Bloch periodicity. DDR has not been formulated with Bloch boundary conditions. Mimetic FD does not handle unstructured meshes. Our construction fills this gap: it restores the discrete exact sequence $d_1\, d_0 = 0$ within the DEC framework for Bloch-periodic problems on general polyhedral complexes. We do not address the full FEEC program (commuting projections, approximation theory); the present contribution is limited to the algebraic exactness of the discrete complex. To date, Bloch-periodic band structure computations on foam-like meshes have used plane-wave expansion [16] rather than DEC, so the exactness defect identified here is latent—it would affect any future DEC-based computation on unstructured periodic complexes, but existing published results are not invalidated.

Limitations. The Hodge stars are diagonal (circumcentric dual); non-diagonal stars would require modification of the assembly but not the exactness construction. The mesh must have valid Voronoi structure; for random point sets, sufficient density is required (≥ 50 points per periodic box gives $> 80\%$ valid Voronoi tessellations; the remaining cases fail at the mesh generation stage due to degenerate dual geometry, not at the exactness construction). We treat only the 3D curl–curl problem; extension to 2D and to time-domain

DEC is straightforward. The construction is lowest-order (piecewise-constant cochains, analogous to Whitney 0-forms); extension to higher-order DEC, where basis functions span more than one cell, would require a corresponding generalization of the face-boundary recurrence. The observed second-order convergence rate (Table 3) is consistent with a lowest-order discretization but a formal error estimate relating mesh size to eigenvalue error is not attempted.

8. Conclusion

The standard Bloch-periodic extension of DEC does not preserve the discrete exact sequence on unstructured polyhedral meshes. This structural inconsistency produces spectral pollution that enters the physical spectrum, compromises the discrete Hodge decomposition, and mesh convergence of the acoustic wave speed is not observed under refinement in the tested regime. A face-boundary recurrence restores exactness by deriving the curl phases from those already present in the gradient, producing a canonical curl–curl operator with correct gauge kernel and acoustic spectrum. The failure is algebraic—it is independent of mesh quality, resolution, or symmetry—and persists under refinement. Verification on simple cubic, Voronoi foam, and random Voronoi complexes confirms that the construction is universal and robust.

Reproducibility

All computations use Python 3.9, NumPy, SciPy (LAPACK/BLAS). Source code and a structured test suite are available at `https://github.com/alex-toader/st_bloch_exactness`. The test suite covers exactness verification, spectral pollution counts, kernel dimension, $\mathbf{k} \to 0$ asymptotics, random $\mathbf{k}$-directions, Brillouin zone boundary behavior, mesh refinement convergence, random Voronoi topologies, operator perturbation norms, full Hodge Laplacian splitting, and face ordering independence. All numerical claims in this paper are backed by reproducible scripts.

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